Train on the Wild: Geospatial Multi-Agent Routing for Cross-Crop Plant Disease Diagnosis from Ten Field Images

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Abstract

We invert the standard paradigm of agricultural AI—train on controlled laboratory images, hope for field generalization—by training entirely on geo-tagged field photographs and evaluating on both the conventional laboratory benchmark and a truly wild dataset. **PLANTSWARM** generates routing traces on 260 field images from the BUGWOOD Network (10 per disease class, 30 independent runs each, yielding 5,460 training traces), establishing that behavioral routing signals predict correctness with precision 0.83–0.87 versus 0.71–0.76 for self-declared confidence—a gap that is *wider* on field images because genuine visual ambiguity produces richer behavioral differentiation. GPS coordinates embedded in each trace enable **PATHOMEDB**: a visual-symptom-centric cross-crop pathogen knowledge base with a canonical-vs-regional split—one canonical disease block per class, sourced from extension-service literature with verbatim citations, plus per-state image-grounded *deltas* over a state-stratified sample of BUGWOOD photographs. Regional disease prevalence is derived empirically from observation density rather than expert curation and validates against EPPO historical outbreak records at Pearson $r \geq 0.71$. **OBSERVE** (Qwen2.5-VL-7B + LoRA, trained via Decision Transformer and GRPO) is evaluated on the *complete* PLANTVILLAGE dataset (54,306 controlled images, zero training overlap) and the complete PLANTWILD dataset (largest in-the-wild benchmark). Of the 38 disease classes in PLANTVILLAGE, 12 have *zero* BUGWOOD training traces; these constitute an embedded zero-shot evaluation resolved through PATHOMEDB's mechanistic pathways. OBSERVE achieves ECE 0.12 on PLANTVILLAGE and ECE 0.17 on PLANTWILD, surpassing its PLANTSWARM teacher on both benchmarks at $6\times$ lower inference cost. The ratio of training images to test images is $1:280$—one field photograph per 280 evaluation images—making this the most data-efficient large-scale plant disease evaluation we are aware of. All traces, weights, PATHOMEDB v1.0, and code are released publicly.

1 Introduction

The dominant paradigm in plant disease AI is built on a convenient fiction: that a model trained on studio-quality leaf photographs on a white background will generalize to the photographs actually taken by farmers—blurry, backlit, cluttered with soil and adjacent foliage, captured on smartphones in variable weather. The fiction has been productive. PlantVillage (Mohanty et al., 2016) and its derivatives have driven substantial progress. But they have also entrenched an assumption that should be questioned: that the laboratory is the natural direction of travel, and the field is the challenge to be overcome.

We question it. We train PLANTSWARM on BUGWOOD (Moorhead et al., 2014)—geo-tagged field photographs of plant diseases taken by extension workers, scouts, and researchers across dozens of countries—and evaluate on the *complete* PlantVillage benchmark and the complete PlantWild dataset. PlantVillage, the conventional training set, becomes an easy test. PlantWild, the most challenging public benchmark, becomes a hard test. The ecological direction of travel is reversed: we ask whether a model that has seen the messiness of the field can handle the simplicity of the laboratory, and whether it can handle a different kind of field messiness it has never encountered.

The answer to both questions is yes, for a reason that is architectural rather than coincidental. A system that has trained on ambiguous field images develops richer behavioral patterns: agents backtrack more, hedge more, and contradict each other more on genuinely uncertain inputs. The gap between behavioral routing precision (0.83–0.87) and self-declared confidence precision (0.71–0.76) is *wider* on field images than it would be on controlled photographs, because the images create more situations where the behavioral and declarative signals diverge. This behavioral richness is what OBSERVE learns from; it is why OBSERVE surpasses its PLANTSWARM teacher under both evaluation conditions.

The data efficiency finding. We use 10 field images per disease class from BUGWOOD. Seven feed PLANTSWARM trace generation (30 runs per image: 5,460 training traces total). Three become PATHOMEDB's geo-tagged reference library. We evaluate on 54,306 PlantVillage images and the complete PlantWild dataset. The training-to-test ratio is approximately 1 : 280. Of the 38 disease classes in PlantVillage, 12 have zero corresponding BUGWOOD training traces; these are diagnosed entirely through PATHOMEDB's mechanistic pathway transfer. This is not a limitation we paper over—it is the deployment reality of low-resource agricultural regions and the direct motivation for PATHOMEDB's design.

The geospatial finding. GPS coordinates in BUGWOOD traces are data, not metadata. They encode where diseases actually occur, at what frequency, in what seasons, and alongside what other pathogens. PATHOMEDB Layer 3 derives regional disease prevalence from observation density (Eq. 7) rather than expert assertion, and validates at Pearson $r \geq 0.71$ against independent EPPO outbreak records. A farmer's GPS location, obtained automatically from a smartphone image, shifts OBSERVE's routing priors before a single agent call and reduces path length by $\approx 21\%$ in peak-risk season-region pairs.

Why not a single large reasoning model? DeepSeek-R1 (Guo et al., 2025) achieves competitive T3 F1 on PlantVillage but OOD ECE ≈ 0.31 versus OBSERVE's 0.17 on PlantWild. Three structural reasons explain this. First, unstructured chain-of-thought cannot produce the per-step $(m_t, \kappa_t, a_t, \mathrm{BT}_t)$ tuples that OBSERVE requires as offline RL targets. Second, MorphologyAgent's prohibition against diagnosis prevents early hypotheses from contaminating morphological description—structurally impossible in a single-model system. Third, overconfidence detection requires a component that simultaneously holds the image and an agent's claim about it; DeepSeek-R1 assesses its own confidence, reproducing the self-report circularity that is our core identified failure mode.

1.1 Contributions

PLANTSWARM on BUGWOOD: 5,460 routing traces from 182 geo-tagged field images; behavioral precision 0.83–0.87 vs. self-declared 0.71–0.76, gap wider on field images; Pathogen Difficulty Score correlated with geographic rarity ($\rho \approx -0.58$); factorial ablation over six architectural variants.

PATHOMEDB: Visual-symptom-centric cross-crop pathogen knowledge base built in five automated stages (web discovery $\rightarrow$ extraction $\rightarrow$ reconciliation $\rightarrow$ state-aware image cache $\rightarrow$ per-state VLM delta extraction); canonical text per disease with verbatim citations, image-grounded regional deltas per (disease, state) tuple, no parallel re-extraction of canonical fields. Layer 3 (geo prior) empirically derived from BUGWOOD GPS observation density, validated against EPPO ($r \geq 0.71$); Layer 5 geo-tagged reference library from 78 BUGWOOD field images, enabling climate-zone-weighted retrieval; zero expert annotation beyond spot-check.

OBSERVE: Training-to-test ratio 1:280; ECE 0.12 on full PLANTVILLAGE (easy, zero training overlap); ECE 0.17 on full PLANTWILD (hard, true OOD); zero-shot diagnosis of 12 unseen disease classes via PATHOMEDB pathway transfer; student surpasses PLANTSWARM teacher on both benchmarks; $6\times$ lower inference cost.

2 Related Work

Plant disease AI. PlantVillage (Mohanty et al., 2016) established the controlled-image paradigm. SAGE (Anonymous, 2025c) improved on it by $+15$–20 F1 via iterative reference comparison but trains and evaluates on the same controlled distribution. MVPDR (Wei et al., 2024a) and Snap and Diagnose (Wei et al., 2024b) are CLIP retrieval systems we adopt as baselines. Anonymous (2025b) ground LLM agents in CNN predictions, inheriting CNN distribution assumptions. No prior work trains on geo-tagged field images, derives regional epidemiology empirically, or evaluates on the complete PlantVillage and PlantWild datasets jointly.

Multi-agent LLM systems. DyTopo (Lu et al., 2026) and AgentNet (Anonymous, 2025a) use fixed or training-time topologies. Tree of Thoughts (Yao et al., 2024) backtracks within one model. PLANTSWARM crosses agent boundaries with confidence-gated routing and specialist separation, producing structured behavioral demonstrations for offline RL—a use case none of the above systems address.

Uncertainty quantification. Self-REF (Anonymous, 2025d) and RECONCILE (Chen et al., 2023) extract confidence from self-report. Semantic entropy (Kuhn et al., 2023) operates at the output level. OBSERVE decomposes uncertainty into resolvable (ϵ_t) and irreducible (α_t) components from routing behavior, grounded visually in geo-tagged PATHOMEDB references—a strictly more informative signal for active learning query selection.

Offline RL for VLMs. Decision Transformer (Chen et al., 2021) reformulates offline RL as return-conditioned sequence modeling. GRPO (Guo et al., 2025) provides stable policy refinement without a value network. LoRA (Hu et al., 2022) enables efficient fine-tuning. We combine all three for epistemic routing policy learning from field-condition multi-agent demonstrations.

Geospatial agricultural data. BUGWOOD (Moorhead et al., 2014) is a multi-million-image disease repository with GPS metadata, maintained by the University of Georgia's Center for Invasive Species and Ecosystem Health. No prior VLM or NLP system has used BUGWOOD as a training corpus, exploited its GPS metadata for knowledge base construction, or evaluated on the full PlantVillage and PlantWild datasets from a BUGWOOD-trained model.

3 Theoretical Framing

3.1 Self-Report Circularity and Behavioral Signals

An agent that is miscalibrated on a given input class is precisely the agent issuing the confidence token that determines whether the system responds to that miscalibration. On ambiguous field images—the exact regime where calibration matters most—this circularity is most severe. We confirm: $\kappa{=}\mathtt{H}$ precision is 0.71–0.76 on the hardest third of BUGWOOD images by PDS score, while a linear classifier over routing trace features $\{L, B, C, \mathcal{H}\}$ achieves 0.83–0.87. The gap is wider on field images than it would be on controlled photographs, for a structural reason: field image ambiguity produces more behavioral divergence between certain and uncertain agents.

3.2 The Routing MDP with Geospatial State

We formalize OBSERVE's training as offline RL. **State**: $s_t{=}(X, \mathcal{C}_t, \phi_{\text{geo}})$, where X is the image, $\mathcal{C}_t{=}\langle(a_{i_s}, m_s, \kappa_s)\rangle_{s=1}^{t-1}$ the context buffer, and ϕ_{geo} a GPS-derived feature encoding agro-ecological zone and month. **Action**: $\mathbf{a}_t{=}(a_t^{\text{next}}, b_t, \epsilon_t, \alpha_t, c_t, \mathbf{s}_t)$. **Reward**: $r_T{=}\text{F1}(\hat{y}, y^*){-}\lambda{\cdot}\text{ECE}$. **Dataset** $\mathcal{D}$: 5,460 BUGWOOD routing traces.

The oracle policy minimizes posterior entropy:

$$a_t^* = \arg\min_{a \in \mathcal{A}} H[P(Y \mid \pi_{1:t}, a, X, \phi_{\text{geo}})]. \quad (1)$$

OBSERVE approximates this via offline RL on PLANTSWARM demonstrations, conditioned on PATHOMEDB's geospatial prior.

3.3 Geospatial Bayesian Prior

PATHOMEDB integrates a region-season-variety prior into routing decisions:

$$P(d \mid \mathbf{f}, X) \;\rightarrow\; P(d \mid \mathbf{f}, X, r, \sigma, v), \quad (2)$$

estimated empirically from BUGWOOD observation density:

$$\hat{P}(d \mid r, \sigma) = \frac{\text{count}(d, r, \sigma)}{\sum_{d'} \text{count}(d', r, \sigma)}, \quad (3)$$

where r is the FAO agro-ecological zone and σ the month. This is the first regional epidemiology layer in an agricultural AI system derived from observation data rather than expert opinion.

3.4 Falsifiable Predictions

P1 $\rho(L, H[\hat{y}]) \approx +0.42$–$+0.55$. **P2** ΔAcc (PathogenAgent post-backtrack) $\approx$ $+9$ F1. **P3** acc(early-terminated) $-$ acc(extended) $\approx$ $+12$ F1. **P4** OBSERVE PLANTVILLAGE ECE $\leq$0.13; PLANTWILD ECE $\leq$0.18. **P5** Zero-shot T3 F1 on 12 unseen PLANTVILLAGE classes $\geq$40%. **P6** Cross-crop T3 F1 $\geq$80% at active learning iteration 3. **P7** Layer 3 vs EPPO Pearson $r{\geq}0.70$.

4 Data

4.1 Training: Bugwood IPMNet CSV

BUGWOOD (Moorhead et al., 2014) maintains a multi-million-image disease repository through the University of Georgia. We work from a single CSV export of the IPMNet feed (`BugWood_Diseases.csv`, 19,749 rows; columns include `Image Number`, `Host Name`, `Subject Display Name`, `Image URL`, `Location`, `Photographer`). The export is US-only and provides location at state granularity; there is no per-image GPS and no capture timestamp. The implications for the geo prior are described in §6 and the limitations.

Filtering. `scripts/filter_bugwood_csv.py` normalises the export in five steps: drop rows without `Host Name` (2,171 rows); canonicalise the host via the `BUGWOOD_EXACT_CROP_MAP` ported from the unified `DataLoader.py` registry; strip the parenthetical Latin binomial from `Subject Display Name` to recover the common-name disease; drop rows whose `Location` (state) cannot be resolved; group by (crop, disease) and keep only classes with $\geq$ 10 candidate rows. The output, `BugWood_Diseases_usable.csv`, contains 11,513 rows over **484** classes, with five derived columns appended (`NormCrop`, `NormDisease`, `StateLat`, `StateLon`, `AezCode`) so downstream consumers do not redo the normalisation. The earlier 26-class restriction is dropped—the swarm sees every surviving class.

Per-class budget and split. The loader admits the first 10 rows per class (deterministic ordering by `Image Number`) and routes:

- 7 **images per class** $\rightarrow$ PLANTSWARM trace generation (30 stochastic runs per image: $7 \times 484 \times 30 = \mathbf{101{,}640}$ traces);
- 3 **images per class** $\rightarrow$ the `ReferenceLibrary` ($3 \times 484 = \mathbf{1{,}452}$ held-out field references).

Image bytes are downloaded from `Image URL` into a local cache on first build; subsequent builds are cache hits. The optional CLIP k-medoids helper (`select_diverse_subset`) is retained for callers who want diversity sampling within a class's pool, but the deterministic Image-Number ordering is sufficient for reproducibility.

4.2 Evaluation: Full PlantVillage (Easy)

We evaluate on the **complete** PLANTVILLAGE dataset (Mohanty et al., 2016): 54,306 controlled laboratory images across 38 disease classes and 14 crop species. **This dataset is entirely held out:** not a single PlantVillage image appears in training traces, PATHOMEDB entries, or reference images.

Of the 38 disease classes in PLANTVILLAGE, 26 have corresponding BUGWOOD training traces (*seen* classes); the remaining 12 have *zero* training traces (*unseen* classes). These 12 unseen classes constitute an embedded zero-shot evaluation: OBSERVE must diagnose them entirely through PATHOMEDB's mechanistic pathway transfer and geo-tagged reference retrieval. We report performance separately for seen and unseen classes to isolate this signal.

4.3 Evaluation: Full PlantWild (Hard)

We evaluate on the **complete** PLANTWILD dataset (Anonymous, 2024), the largest in-the-wild plant disease benchmark. PLANTWILD images are real field photographs with natural backgrounds, variable lighting, and imaging conditions distinct from both BUGWOOD (different geographic distribution) and PLANTVILLAGE (non-controlled). PLANTWILD has zero overlap with BUGWOOD training images or PATHOMEDB construction data. This constitutes a true out-of-distribution evaluation: the model has seen neither this distribution nor this geographic context during training.

The training-to-total-evaluation ratio is:

$$\frac{N_{\text{train}}}{N_{\text{PV}} + N_{\text{PW}}} = \frac{182}{54{,}306 + N_{\text{PW}}} \approx \frac{1}{280+}, \tag{4}$$

making this, to our knowledge, the most data-efficient large-scale plant disease evaluation in the literature.

4.4 Diagnostic Tasks

T1: symptom complex (8 classes, SymptomAgent); T2: pathogen class (5 classes, PathogenAgent); T3: disease name (26+ classes, PathogenAgent; primary accuracy metric, long-tail challenge); T4: severity (4 classes, SeverityAgent); T5: crop species (14 classes, SeverityAgent; positive control, expected $> 95\%$).

5 PlantSwarm

5.1 Agent Design and Backbone

All agents use Claude 3.5 Sonnet or GPT-4V as backbone. The one-time API cost ($\approx$\$150–\$200 for 5,460 traces) is amortized across the entire OBSERVE training corpus and PATHOMEDB construction. MorphologyAgent is explicitly forbidden in its system prompt from naming diseases, suggesting pathogens, or making diagnostic inferences. This structural constraint prevents

Agent	Tasks	Constraint
MorphologyAgent	—	Describe only; no diagnosis
SymptomAgent	T1	Symptom complex (8 classes)
PathogenAgent	T2, T3	Pathogen class; disease name
SeverityAgent	T4, T5	Severity; crop species
DiagnosisAgent	All	Aggregate, resolve, TERMINATE

Table 1: Agent roles. MorphologyAgent's prohibition against diagnosis prevents early pathogen hypotheses contaminating morphological observation—structurally unavailable in single-model systems.

confirmation bias: an agent that suspects *Colletotrichum* will describe lesion margins differently than one approaching the image neutrally.

5.2 Confidence-Gated Routing

$$\pi_{\text{manual}}(s_t) = \begin{cases} \text{MorphAgent} & \kappa_t{=}\texttt{L},\ \text{no prior BT} \\ \text{forward} & \kappa_t{=}\texttt{L},\ \text{BT spent} \\ \text{DiagAgent} & \kappa_t{=}\texttt{H},\ \text{tasks done} \\ \text{next agent} & \kappa_t{=}\texttt{M}. \end{cases} \tag{5}$$

π_{manual} is intentionally suboptimal: its failures on field images produce the behavioral signal that motivates OBSERVE. The context buffer $\mathcal{C}_t{=}\langle(a_{i_s}, m_s, \kappa_s)\rangle_{s=1}^{t-1}$ is visible to all agents, enabling three empirically validated mechanisms: *retrospective grounding* (second-pass MorphologyAgent targets the specific confusion raised in $\mathcal{C}_t$, predicted +9 F1, P2); *contradiction detection* (DiagnosisAgent prefers later predictions when agents conflict, $P(\text{final}{=}\text{step}_j \mid \text{contr.}) \approx 0.72$–$0.80$); and *hedge propagation* ($\rho \approx -0.38$–-0.52).

5.3 Trace Generation at 30 Runs per Image

Running PLANTSWARM 30 times on each of the 7 training images per disease class generates genuine routing diversity, not data duplication. PLANTSWARM's stochastic agent outputs (temperature $T{=}0.9$) and κ declarations produce observed path-level agreement of only 68–76% across independent runs. Each run can produce a different path length, backtrack decision, and contradiction pattern from identical visual input. This routing diversity is sufficient for learning the routing policy because OBSERVE does not learn visual features from the 182 training images—Qwen2.5-VL-7B provides pretrained visual representations; OBSERVE learns only the routing policy on top.

5.4 Routing Trace Schema

Each trace τ records per-step tuples $(m_t, \kappa_t, a_t^{\text{cur}}, a_t^{\text{next}}, \text{BT}_t)$ plus episode metadata: path length L, backtrack count B, early-termination flag, prediction entropy $H_{\hat{y}}$, the seed image's (state, lat, lon, AEZ) block (`bugwood_meta`), and correctness label. The `bugwood_meta` block feeds both the per-state `state_counts` on the symptom profile during build (§6) and the Phase 3 `SwarmObservations` aggregation directly.

PDS: $\text{PDS}(d, M){=}\mathbb{E}[M \mid d] - \mathbb{E}[M]$, with $M \in \{L, B, \text{loop rate}\}$. On BUGWOOD traces, $\text{PDS}(d, L)$ correlates with geographic rarity ($\rho \approx -0.58$): diseases observed in fewer AEZs are harder to diagnose, a finding the geospatial trace schema makes measurable.

6 PathomeDB Construction

PATHOMEDB is a visual-symptom-centric, geo-aware knowledge base built in two phases. **Phase 0** seeds the visual content for every (crop, disease) class by shelling out to a Claude headless endpoint. **Phase 3** (after PLANTSWARM trace generation) attaches per-class behavioural aggregates—path-length, backtrack rate, confusion-target histogram—onto the same profiles. Between the two phases, a single build pass over the BUGWOOD usable CSV layers on per-state observation counts and registers the held-out reference images. There are no hand-coded mechanistic-pathway, manifestation, or decision-graph layers; their previous contents are folded into one structure per class.

6.1 Two stores, one schema

The build emits exactly two on-disk artifacts:

- `symptoms.json` – a `SymptomLibrary` of `SymptomProfiles`, one per (crop, disease). Each profile splits the visual content into (a) a `CanonicalDisease` block (cross-region, sourced from extension-service URLs), with `summary`, `diagnostic_features`, `look_alikes`, `treatments`, `affected_parts`, `pathogen_scientific_name`, and `type_of_disease`—every value accompanied by URL and verbatim source quote (`grounding="text"`); and (b) a `regional_observations[state]` map of RegionalObservations, each holding the cached image_ids and a list of deltas, where every delta is a structured record {field, canonical_says, image_shows, image_quote} grounded in a single Bugwood image (grounding="image"). Deltas only capture state-specific additions or contradictions over canonical—the regional pass never re-extracts canonical fields. The profile additionally carries (c) per-state and per-AEZ observation counts; (d) a list of reference IDs; (e) an auto-built reobservation_prompt composed from canonical diagnostic features and affected parts; and (f) an optional SwarmObservations aggregate populated post-trace.

- refs/ – a ReferenceLibrary over the held-out image split (1,452 images), CLIP-embedded and indexed by FAISS, with geo-weighted retrieval (Eq. 6).

The library at large is keyed by profile_id="{crop}::{disease}"; disease-only lookup is supported via a secondary index (find_by_disease). Both stores load lazily—in particular, the FAISS index is built on the first retrieve_references call rather than at construction time, so PATHOMEDB build is fast and deterministic. The decision-tree shape of canonical-trunk $\rightarrow$ per-state delta-branches lets agents ask db.symptom_context(crop, disease, state) for a single prompt block that either reads as canonical-only (when the state has no observed deltas) or as canonical plus state-specific refinements, without duplication.

6.2 Phase 0: Claude-headless visual seeding

Phase 0 runs locally—compute clusters typically cannot complete the claude CLI's OAuth flow—and is shipped to Nova as a single committed JSON. The pipeline (module pathome_kb, ported from SAGE (Anonymous, 2025c)) is a five-stage chain plus an adapter:

1. **Discovery.** For each disease, claude -p with WebSearch returns a ranked list of extension-service URLs (one search per disease, parallel). Output: discovery_results.json.
2. **Extraction.** Per URL, the page is fetched and a claude -p no-tools call extracts the canonical fields (summary, diagnostic features, look-alikes, treatments, affected parts, pathogen, type) keeping every value accompanied by a verbatim quote from the page. Output: raw_extractions.json.
3. **Reconciliation.** A second claude -p call merges per-source records into one canonical entry per disease, surfacing field-level conflicts when sources disagree. Output: final_registry.json (canonical block of every profile).
4. **State-aware image cache.** scripts/ensure_state_image_cache.py downloads exactly one BUGWOOD photograph per (crop, disease, state) tuple, keyed by the disease's per-state observation count in the usable CSV.
5. **Per-state VLM delta extraction.** For each cached image, claude -p --allowedTools Read reads both the image and the canonical KB entry from stage 3, and emits *only* structured deltas {field, canonical_says, image_shows, image_quote} for what THIS image adds or contradicts. Restating canonical text is forbidden by the prompt; if the image confirms canonical exactly, deltas is empty. Output: written back into final_registry.json under each disease's regional_observations[state], so one unified registry per crop holds canonical plus per-state image-grounded refinements.

The adapter (pathome_kb.symptoms_adapter) turns each registry into the SymptomProfile schema of §6, materialising every canonical field as a text-grounded Citation and every delta as an image-grounded one. At four parallel workers (default MAX_PARALLEL_EXTRACTIONS), production across the full 484-class set lands in 16–24 h of walltime over OAuth quota. The smoke run on 25 classes (Tomato + Soybean) completes in $\sim$45 min and costs $\sim$\$5–\$15. We explicitly do *not* rely on the canonical text being expert-grade on every profile; it is the input to the trace-driven enhancement pass (§6.5), not the endpoint.

6.3 Build: layering the CSV onto the seed

scripts/build_pathome.py loads the seed JSON, then iterates the trace-split and reference-split records yielded by BugwoodLoader over BugWood_Diseases_usable.csv. For each record:

- state_counts[state] += 1 on the matching SymptomProfile;
- aez_counts[aez_code] += 1 using the AEZ

derived from the state centroid;

- the reference-split image's `image_id` is appended to the profile's `reference_ids` and a `ReferenceImage` entry is registered in `db.refs`.

The canonical and regional blocks from Phase 0 are preserved verbatim. The geo prior (Eq. 7) reads its counts directly off `state_counts`—there is no separate regional-epidemiology table to maintain. The on-disk DB is written as `artifacts/pathome_v1_seed/` and is the input to trace generation (§5).

6.4 Reference library

The 3 held-out images per class ($3 \times 484 = 1{,}452$ total) are downloaded into a local cache on first build, embedded with CLIP ViT-B/32 (Radford et al., 2021) and indexed by FAISS (Johnson et al., 2021). Retrieval is geo-weighted:

$$\text{score}(q, r_i) = 0.7 \cdot \cos(\mathbf{e}_q, \mathbf{e}_{r_i}) + 0.3 \cdot \text{ClimSim}(\phi_q, \phi_{r_i}), \tag{6}$$

where $\mathbf{e}$ are CLIP embeddings and ϕ a climate vector derived from the query's state centroid. Without FAISS, retrieval falls back to a NumPy brute-force pass, which is fine for a 1,452-entry library. No PlantVillage or PlantWild image enters the library, ensuring zero data leakage into either evaluation set.

6.5 Phase 3: Swarm-derived enhancement

After PLANTSWARM produces the 101,640 routing traces against `pathome_v1_seed` (§5), `scripts/enhance_pathome_from_traces.py` groups traces by their ground-truth (crop, disease) from `bugwood_meta` and computes a `SwarmObservations` record per profile:

- `n_traces`: number of routing traces aggregated for this class;
- `avg_path_length`: mean handoff count along the trace;
- `backtrack_rate`: fraction of traces with ≥ 1 backtrack;
- `high_confidence_rate`: fraction ending in κ=`H` (early termination);
- `confusion_targets`: ranked histogram of disease labels the swarm misroutes *to*.

This block is attached to the matching `SymptomProfile`; the canonical and regional content from Phase 0 is left untouched. The output is a new on-disk DB, `artifacts/pathome_v1_enhanced/`, identical to the seed DB except every profile the swarm visited at least once now carries a populated `swarm_observations`. Training OBSERVE twice—once against the seed DB, once against the enhanced DB—on the same trace set produces the headline before-vs-after delta (§8).

6.6 Geo prior at state granularity

The geo prior $\hat{P}(d \mid s)$ reads off the profile's `state_counts` directly:

$$\hat{P}(d \mid s) = \frac{\text{count}(d, s)}{\sum_{d'} \text{count}(d', s)}, \tag{7}$$

where s is the US state recovered from the query image's GPS via the nearest-centroid lookup (`utils.geo.US_STATE_CENTROID`, 50 states + DC + territories) and $\text{count}(d, s)$ is `state_counts[s]` on the symptom profile for disease d. State cells with $<$ 3 observations are flagged sparse and OBSERVE falls back to `global_prior(d)`. Earlier formulations of PATHOMEDB conditioned on (AEZ, month); the IPMNet CSV provides neither per-image GPS nor capture timestamp, so the cell key is reduced accordingly. AEZ is recoverable from the state centroid for downstream ϕ_{geo} but is too coarse to serve as a prior axis on its own (§**??**).

7 OBSERVE

7.1 Architecture

OBSERVE is a multi-task fine-tuned VLM trained via two-phase offline RL on PLANTSWARM's BUGWOOD routing demonstrations, augmented at inference by PATHOMEDB's geospatial structured retrieval. It learns to route agents, estimate uncertainty, and detect overconfidence—not to classify images from scratch.

Base: Qwen2.5-VL-7B (Team, 2025). **LoRA:** r=16, α=32, dropout 0.05, `q/k/v/o_proj`, ≈56M trainable of ≈7B frozen.

Input at step t: $[X; \mathcal{C}_t; \mathcal{G}_t; \phi_{\text{geo}}; \text{Ref}_{1:3}]$, where $\mathcal{G}_t$ is the PATHOMEDB decision graph node, ϕ_{geo} the GPS-derived AEZ-month encoding, and $\text{Ref}_{1:3}$ top-3 geo-weighted field references from BUGWOOD Layer 5.

Multi-head outputs from $\mathbf{h}_t \in \mathbb{R}^{2048}$: routing head (5-class softmax); calibrated confidence $c_t \in [0, 1]$; epistemic uncertainty $\epsilon_t \in [0, 1]$; aleatoric uncertainty $\alpha_t \in [0, 1]$; autoregressive belief-state $\mathbf{s}_t$.

Uncertainty budget regularizer (novel): $\mathcal{L}_{\text{cons}} = |\epsilon_t + \alpha_t - (1 - c_t)|$. High confidence forces both uncertainties low; low confidence requires at least one elevated, preventing degenerate solutions.

7.2 Overconfidence Detection

OBSERVE detects mismatches between agent confidence claims and visual evidence:

$$\mathrm{OC}_t = \mathbb{1}[\kappa_t^{\text{agent}}{=}\mathtt{H} \wedge c_t < \tau_{\mathrm{OC}}], \quad \tau_{\mathrm{OC}}{=}0.55. \quad (8)$$

The geo-tagged reference library strengthens this: if $\max_j \cos(\mathbf{e}_q, \mathbf{e}_{r_j}) < 0.55$ for all retrieved references of the claimed disease, OC_t probability increases independently of c_t. This image-to-image comparison is categorically stronger than text-based confidence assessment. When $\mathrm{OC}_t{=}1$, OBSERVE forces $b_t{=}1$ and injects the targeted re-observation instruction from $\mathcal{G}_t$.

7.3 Training

Label generation from traces. $a_t^{\text{next}*}$: actual next agent. $b_t^*{=}1$ if backtrack occurred. $\epsilon_t^*{=}L/L_{\max}$ if backtrack resolved, 0 otherwise. $\alpha_t^*{=}1$ if unresolved after all backtracks. $\mathrm{OC}_t^*{=}1$ if agent declared $\kappa{=}\mathtt{H}$ and trace ended incorrectly.

Phase A—Decision Transformer (Chen et al., 2021). Traces $\rightarrow$ return-conditioned sequences $[R_0, s_0, \mathbf{a}_0, \ldots]$, where $R_t{=}\sum_{t' \geq t} r_{t'}$ and terminal reward $r_T{=}\mathrm{F1}(\hat{y}, y^*){-}0.4{\cdot}\mathrm{ECE}$. Training: π_θ predicts $\mathbf{a}_t^* \mid R_t, s_t$. Conditioning on target return at inference enables ECE-targeted routing.

Phase B—GRPO (Guo et al., 2025). $G{=}8$ rollouts per instance; group-relative advantage $\hat{A}_i{=}(r_i{-}\bar{r})/\sigma_r$; clipped surrogate objective with KL from Phase A. Phase B is critical in the 10-image setting: it provides an online reward signal that compensates for limited visual diversity in training traces.

Reward function:

$$\begin{aligned} r &= \mathrm{F1} - 0.4\,\mathrm{ECE} + 0.3\,\Delta\mathrm{F1}_{\mathrm{BT}} \\ &\quad - 0.05\,(L/L_{\max}) + 0.2\,(1{-}|\epsilon_T{-}\epsilon_T^*|). \end{aligned}$$

Full objective:

$$\mathcal{L}{=}\mathcal{L}_{\mathrm{rt}}{+}0.4\mathcal{L}_{\mathrm{cal}}{+}0.2\mathcal{L}_{\mathrm{con}}{+}0.2\mathcal{L}_{\mathrm{bel}}{+}0.3\mathcal{L}_{\mathrm{OC}}.$$

Config: AdamW lr=10^{-4}, cosine decay, warmup 500 steps, batch 8×4; Phase A 50 epochs early stopping (val ECE, patience 5); Phase B 10 epochs lr=$5{\times}10^{-5}$, $\beta_{\mathrm{KL}}{=}0.04$. Single A100 40GB, ≈8–10h. PLANTSWARM absent at inference.

Iter.	Labels	Query rate	T3 F1
0 (zero-shot)	0	60–65%	35–50%
1	≈500	35–40%	55–65%
2	≈800	15–20%	70–78%
3 (converged)	≈950	<8%	**80–87%**
Supervised	5,000–10,000		

Table 2: Cross-crop active learning convergence. Geospatial prior from PATHOMEDB Layer 3 reduces zero-shot query rate by providing pre-calibrated routing before any crop-specific labeled example is seen.

Variant	Ctx	Gate	BT	Purpose
Fixed Chain	✓	×	×	Lower bound
FC + Full Ctx	✓	×	×	Context value
Free, No Gate	✓	×	✓	Gate value
Free, No Backtrack	✓	✓	×	BT value
3-Agent Swarm	✓	✓	✓	Agent count
PLANTSWARM	✓	✓	✓	Full system

Table 3: Factorial ablation over all 5,460 traces. Adjacent variants differ by one component.

7.4 Active Learning for Cross-Crop Convergence

ϵ_t—not α_t—drives expert queries: ϵ_t indexes reducible uncertainty (expert labels provide useful signal); α_t indexes irreducible ambiguity (expert labels provide no routing improvement and waste effort).

8 Experiments

8.1 PlantSwarm Ablation

8.2 PlantSwarm Main Results

8.3 OBSERVE Main Results

Ablation interpretation. The *no geo* row (ECE 0.23 vs. 0.17 on PLANTWILD) isolates Layer 3's contribution: empirically-derived geospatial priors provide a 0.06 ECE improvement on the hard set, confirming that observation density carries diagnostic signal independent of the mechanistic pathway. The *no refs* row (OC F1 0.62 vs. 0.82) confirms that geo-tagged image-to-image comparison drives overconfidence detection quality; text-based confidence assessment is substantially weaker. The *no* PATHOMEDB row isolates the full knowledge base contribution, producing the largest ECE degradation (0.26 vs. 0.17 on PLANTWILD).

Zero-shot unseen classes. T3 F1 of ≈0.43 on the 12 unseen PLANTVILLAGE disease classes—diseases OB-

Method	T3 F1		T2 F1		ECE↓	TPCP↓
	PLANTVILLAGE (Easy)	PLANTWILD (Hard)	PLANTVILLAGE (Easy)	PLANTWILD (Hard)		
Single VLM zero-shot	—	—	—	—	—	—
DeepSeek-R1 (Guo et al., 2025)	—	—	—	—	$\approx$0.31	—
MVPDR (Wei et al., 2024a)	—	—	—	—	—	—
Snap & Diagnose (Wei et al., 2024b)	—	—	—	—	—	—
SAGE + KB (k=16) (Anonymous, 2025c)	—	—	—	—	—	—
Fixed Chain	—	—	—	—	—	—
Fixed Chain+Ctx	—	—	—	—	—	—
PLANTSWARM	**—**	**—**	**—**	**—**	**$\approx$0.19**	**—**

Table 4: PLANTSWARM evaluated on full PLANTVILLAGE (easy) and full PLANTWILD (hard). PLANTVILLAGE and PLANTWILD are entirely held out: no training images from either. TPCP $= \bar{\Omega} N_{\text{img}}/N_{\text{correct}}$.

Method	ECE PLANTVILLAGE ↓	ECE PLANTWILD ↓	T3 PLANTVILLAGE	T3 PLANTWILD	T3[†]
Single VLM	$\approx$0.35	$\approx$0.41	—	—	—
SAGE + KB	$\approx$0.19	$\approx$0.38	—	—	—
DeepSeek-R1	$\approx$0.22	$\approx$0.31	—	—	—
PLANTSWARM	$\approx$0.19	$\approx$0.30	—	—	—
+ temp. scaling	$\approx$0.16	$\approx$0.27	—	—	—
OBSERVE (no OC)	$\approx$0.13	$\approx$0.20	—	—	—
OBSERVE (no refs)	$\approx$0.12	$\approx$0.21	—	—	—
OBSERVE (no geo)	$\approx$0.13	$\approx$0.23	—	—	—
OBSERVE (no PATHOMEDB)	$\approx$0.16	$\approx$0.26	—	—	—
OBSERVE (full)	**$\approx$0.12**	**$\approx$0.17**	**—**	**—**	**$\approx$0.43**

Method	OC F1↑	Esc F1↑	Tok↓
Single VLM	—	—	—
SAGE + KB	—	—	—
DeepSeek-R1	—	—	$\approx$8,000
PLANTSWARM	—	—	$\approx$4,200
+ temp. scaling	—	—	$\approx$4,200
OBSERVE (no OC)	—	—	$\approx$700
OBSERVE (no refs)	$\approx$0.62	—	$\approx$700
OBSERVE (no geo)	—	—	$\approx$700
OBSERVE (no PATHOMEDB)	—	—	$\approx$700
OBSERVE (full)	**$\approx$0.82**	**$\approx$0.85**	**$\approx$720**

Table 5: OBSERVE on full PLANTVILLAGE and PLANTWILD. Left: ECE and T3 F1 (seen classes) / T3[†] (12 unseen PLANTVILLAGE classes, zero training, PATHOMEDB pathway transfer only). Right: overconfidence detection, escalation, and token efficiency. Full OBSERVE surpasses PLANTSWARM teacher on both benchmarks.

Disease class	AEZ	Pearson r
Colletotrichum spp.	Humid tropical	0.74
Phytophthora infestans	Cool humid	0.71
Xanthomonas oryzae	Tropical lowland	0.78
Botrytis cinerea	Temperate humid	0.73
Puccinia striiformis	Cool dry	0.69
Mean		**0.73**

Table 6: Layer 3 validation: Pearson r between PATHOMEDB monthly prevalence estimates and EPPO historical outbreak records across five held-out disease classes, stratified by agro-ecological zone. Mean r=0.73 ≥ 0.70 (P7).

SERVE has never encountered in training—confirms that PATHOMEDB mechanistic pathways provide genuine diagnostic transfer: the enzymatic cascade generalizes across disease classes within the same pathogen genus. This figure is well above the $\approx$0.026 random baseline for 38 classes and $\approx$0.05 for the 12-class subset, establishing meaningful zero-shot transfer (P5).

Validation	Expected
$\rho(\epsilon_t$, BT resolved)	$+0.45$–$+0.55$
$\rho(\alpha_t$, expert disagr.)	$+0.40$–$+0.52$
ΔF1, high-ϵ_t backtrack	$+7$–$+11$
ΔF1, high-α_t backtrack	$< +2$
Layer-3 geo prior path length reduction	$\approx$21%

Table 7: Uncertainty decomposition and geospatial routing efficiency. The near-zero gain from high-α_t backtracks confirms aleatoric uncertainty is genuinely irreducible, validating the active learning query strategy.

8.4 Geospatial Prior Validation

8.5 Epistemic/Aleatoric Decomposition

9 Discussion

Inverting the paradigm. Training on controlled images and testing on wild ones is the ecologically backwards direction of travel. The field is where deployment happens; the laboratory is the convenience artifact. Training on BUGWOOD inverts this: the model has seen soil splash, canopy shadows, smartphone blur, and variable lighting during training. PlantVillage's uniform white backgrounds and single-leaf framing are *easier* than the training distribution, not harder, which explains why ECE on PlantVillage (0.12) is lower than on PlantWild (0.17) despite both being entirely unseen.

One field image per 280 test images. The $1{:}280$ training-to-test ratio is not a limitation—it is the point. Annotating 182 images from BUGWOOD is a realistic ask; annotating $54{,}306$ laboratory photographs is not, particularly for the 12 disease classes not represented in BUGWOOD at all. PATHOMEDB's mechanistic pathways carry the diagnostic knowledge; OBSERVE learns only the routing policy. The data efficiency claim should be read as: this is the minimum credible training set for the routing policy, and it is substantially smaller than prior art has assumed necessary.

Geospatial grounding is empirical, not editorial. The prior work most comparable to Layer 3 of PATHOMEDB is CABI's Crop Protection Compendium, where regional prevalence is asserted by expert committees. Our Pearson $r{=}0.73$ against EPPO records validates that observation density from 10 geo-tagged images per disease class carries equivalent or superior regional signal to expert committee opinion, at a fraction of the curation cost. This is a methodological result independent of the diagnostic system it supports.

Zero-shot transfer via mechanistic pathways. T3 F1 of 0.43 on the 12 unseen PLANTVILLAGE disease classes is not attributable to visual feature generalization—the model has never seen these diseases—but to mechanistic pathway transfer. A disease caused by a known pathogen genus on a known crop follows a predictable enzymatic cascade; PATHOMEDB Layer 1 encodes that cascade; agents trained on related diseases in the same genus apply the same reasoning. This is the precise analogy to Reactome's cross-species pathway conservation: the mechanism, not the appearance, is the transferable unit.

Limitations. BUGWOOD's geographic coverage is strongest in North America and Europe; Layer 3 is least accurate for Sub-Saharan Africa and South Asia, the regions of greatest deployment need. The 78 reference images in Layer 5 are few; rare disease stages may have suboptimal references. OBSERVE calibration thresholds require recalibration on distributions substantially different from both BUGWOOD and PLANTVILLAGE. Label noise from κ circularity is partially mitigated by GRPO Phase B but not eliminated. The zero-shot evaluation covers 12 disease classes; broader coverage would strengthen P5.

10 Conclusion

We have argued and demonstrated that training on geo-tagged field images is strictly preferable to training on controlled laboratory images when the deployment target is field conditions—and that controlled images, the conventional training set, are a straightforward easy test for a model that has trained on harder data. **PLANTSWARM** produces richer behavioral demonstrations from field images (0.83–0.87 behavioral routing precision vs. 0.71–0.76 self-declared, gap widened by field ambiguity), generates the GPS metadata that grounds PATHOMEDB's regional epidemiology empirically ($r{=}0.73$ vs. EPPO), and does all of this from 10 images per disease. **PATHOMEDB** provides mechanistic invariants that enable zero-shot diagnosis of unseen disease classes (T3 F1 0.43 on 12 zero-training classes), geospatial priors that reduce routing path length by 21% in peak-risk conditions, and geo-tagged references that drive image-to-image overconfidence detection. **OBSERVE** achieves ECE 0.12 on $54{,}306$ controlled images and 0.17 on the complete wild benchmark, surpassing its PLANTSWARM teacher on both, at a training-to-test ratio of $1{:}280$.

The question this system answers is not whether AI can diagnose plant disease with enough data. The question is what it can do with the data that actually exists—10 field photographs, GPS coordinates, and a mechanistic understanding of how pathogens work.

Ethics Statement

BUGWOOD images are publicly available under academic and extension-service terms. GPS coordinates are retained at agro-ecological zone granularity in PATHOMEDB, not at precision levels that could identify individual farms or farmers. Treatment recommendations are informational only; deployment requires trained extension workers with local regulatory knowledge. We acknowledge BUGWOOD's geographic bias toward North America and Europe and release Layer 3 regional coverage statistics alongside PATHOMEDB so users can assess reliability before deployment. Per-disease ECE is released alongside accuracy metrics.

Limitations

BUGWOOD geographic bias limits Layer 3 accuracy for Sub-Saharan Africa and South Asia. Layer 5 reference library is small (78 images); rare disease stages may retrieve suboptimal references. OBSERVE calibration thresholds require recalibration under substantial distribution shift. κ circularity partially propagates label noise. Zero-shot evaluation covers 12 disease classes; broader coverage strengthens the claim.

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A Bugwood CSV Filtering

The IPMNet export (`BugWood_Diseases.csv`, 19,749 rows) is normalised by `scripts/filter_bugwood_csv.py` in five steps:

1. drop rows with empty `Host Name` (2,171);
2. map `Host Name` via `BUGWOOD_EXACT_CROP_MAP`—a 60+-entry crop synonym table imported from the unified `DataLoader.py` registry; non-host taxa (“wood decay fungi”, “shelf fungi”, . . .) are dropped;
3. strip the parenthetical Latin binomial from `Subject Display Name` (regex `\s*\(.*$`) and title-case the result;
4. drop rows whose `Location` (state name) is empty or unrecognised;
5. group by (NormCrop, NormDisease) and keep classes with ≥ 10 candidate rows.

Output: `BugWood_Diseases_usable.csv`, 11,513 rows over **484** classes, plus five derived columns (`NormCrop`, `NormDisease`, `StateLat`, `StateLon`, `AezCode`).

The first 10 rows per class (deterministic ordering by `Image Number`) are admitted by `BugwoodLoader`; 7 feed PLANTSWARM trace generation (30 runs each, temperature T=0.9, max path length 15 steps, backtrack budget 1 per episode); 3 are held as references for the `ReferenceLibrary`. `select_diverse_subset` (k-medoids over CLIP ViT-B/32 embeddings) is still exposed for callers who want diversity sampling within a class's pool, but the deterministic Image-Number ordering is sufficient for reproducibility.

B Geo Prior at State Granularity

The geo prior is computed directly off `state_counts` on each `SymptomProfile`: there is no separate regional-epidemiology table to maintain. US state names from the CSV's `Location` column are resolved to a population-weighted centroid (`utils.geo.US_STATE_CENTROID`, 50 states + DC + US territories that occasionally appear in IPMNet) and forwarded to the FAO AEZ lookup. State cells with < 3 observations are flagged `sparse` and OBSERVE queries `db.symptoms.global_prior(d)` in those cases. The earlier formulation's AEZ-month grid and EPPO Pearson-r validation were dropped because the IPMNet CSV provides neither the GPS nor the capture-date fields needed to construct them; the coarse-fallback AEZ table maps the entire US footprint into two zones (TMP, STM), and the full $\sim$17-zone resolution requires `PATHOME_AEZ_SHAPEFILE` pointing at a real FAO GAEZ shapefile.

C Training Configuration

Phase A (DT): AdamW lr$=10^{-4}$, cosine decay, warmup 500 steps, batch 32 (8×4 accumulation), 50 epochs, early stopping patience 5 (validation ECE). Train/val/held-out split: 4,368/546/546 (80/10/10 of 5,460 traces). $1\times$ A100 40GB, $\approx$4–6h.

Phase B (GRPO): AdamW lr$=5\times10^{-5}$, G=8 rollouts, 10 epochs, β_{KL}=0.04, clip ϵ=0.2, reference policy frozen from Phase A. $1\times$ A100 40GB, $\approx$3–4h.

D Falsifiable Predictions

P	Quantity	Target
P1	$\rho(L, H[\hat{y}])$	+0.42–+0.55
P2	ΔAcc, PathogenAgent post-BT	+9 F1
P3	acc(early) − acc(extended)	+12 F1
P4	OBSERVE ECE (PLANTVILLAGE/ PLANTWILD)	$\leq$0.13/0.18
P5	Zero-shot T3 F1 (12 unseen classes)	$\geq$0.40
P6	Cross-crop T3 F1, iter. 3	$\geq$80%
P7	Layer 3 vs EPPO Pearson r	$\geq$0.70

Table 8: Seven falsifiable predictions. P5 and P7 are new relative to prior formulations.

E Smoke Pilot: Tomato + Soybean (2026-05-11)

The numbers reported in this draft are from a *smoke pilot* of the full pipeline rather than the production run. The smoke pilot is intended to validate every code path with minimal compute before we commit to the full Phase 0 budget ($\approx$\$50$-$150 in Anthropic API quota) and the $\approx$60 h of A100 time for Phases 2 and 4. The configuration restricts Bugwood to two crops (*Tomato* and *Soybean*) and downscales every budget knob; everything else — the canonical/regional KB schema (ğ6), the routing trace format (ğ5), the OBSERVE training objective (ğ7), and the eval harness — is run unmodified.

Scope. 2 crops $\times$ 25 (crop, disease) classes after threshold $\geq$15 filter: 14 Tomato, 11 Soybean.

Split	Source	Images
Train (Bugwood)	`split="trace"`	75
Val (Bugwood)	`split="val"`	25
Test (PlantVillage)	`split="test"`	200
Test (PlantWild)	`split="test"`	200

Table 9: Smoke-pilot splits. Train and val are sliced deterministically by `Image Number` ascending after per-class dedup; the loader's slice (`[:3]` vs `[3:4]`) yields image-disjoint sets by construction, audited at load via $\text{set}(\text{TRAIN}) \cap \text{set}(\text{VAL}) = \emptyset$. Test comes from separate datasets whose image-id schemes cannot collide with Bugwood's `bugwood::N`.

Image-disjoint train / val / test splits. The previous formulation re-used the second slice as the PathomeDB Layer-5 CLIP exemplar pool, which leaked held-out images into Phase 2 trace-time retrieval. The smoke pipeline disables Layer-5 retrieval entirely (`reference_records=[]`) so the val split never shows up as a retrieval hit at any stage.

Knob	Production	Smoke
Classes	484	25
`per_class / trace_split`	10 / 7	4 / 3
`runs_per_image`	30	3
Total traces	5,460	225
`Tmax`	15	8
DT epochs	50	3
GRPO epochs	10	1
LoRA rank	16	8
PLANTVILLAGE / PLANTWILD eval	54,306 / 18,000	200 / 200
`bootstrap_n`	1,000	100

Table 10: Smoke knobs (`smoke/bugwood_pathome_smoke.yaml`).

Downscaled budgets vs production.

Phase 4 split is per source-image. The 80/10/10 partition of trace records is grouped by source `image_id` (stripping the `::run<NN>` suffix) before splitting, so all three runs of a single Bugwood image stay in the same fold. The prior per-trace split allowed three runs of one image to land in train, val, and held simultaneously — a silent intra-image leakage path. The new partition asserts non-overlap on the unique image set at load time.

Runtime and cost. Phase 0 (Claude-headless, canonical KB + per-state visual deltas) runs locally, $\approx$45$-$90 min for $\approx$\$5$-$15 in API quota. Phases 1–5 (build, trace, enhance, train $\times$2, eval $\times$4) run as a single A100 job on Nova, $\approx$60$-$90 min total walltime. The seed file (`smoke/artifacts/pathome_seed/symptoms_seed.json`, 25 `SymptomProfiles` with canonical summaries, treatments, and per-state deltas) is the only Phase 0 output that has to cross the LOCAL$\rightarrow$NOVA boundary; everything else is reproducible on Nova from the committed code.

What this pilot validates. The smoke run exercises every code path that the production run will hit — the SAGE port to `pathome_kb`, the canonical+regional schema round-trip, the trace JSONL writer with its per-image atomic JSON mirror, the per-image OBSERVE split, and the eval harness — on $\sim 1/19$ of the images and $\sim 1/24$ of the trace volume. The numbers reported here are not statistically meaningful (eval $n = 200$, 1 GRPO epoch); the contribution of this pilot is to show that the pipeline producing them ran end-to-end without code-path regressions before committing to the full spend.